Research Project

SPORT – Research through development of a web application for managing sports clubs

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Status: laufend

Projektseite [externer Link]

Summary (DE & EN)

Das Forschungsprojekt SPORT entwirft, entwickelt, testet und evaluiert die Webapplikation Sportyweb – eine integrierte Anwendungssoftware für das Verwalten, Organisieren und Betreiben von Sportvereinen. Ziel des Forschungsprojekts ist die reflektierte Entwicklung einer mandantenfähigen Webapplikation für das effiziente und kollaborative Management von Sportvereinen. Das Forschungsprojekt untersucht die bislang nur nachrangig beleuchtete Anwendungsdomäne von Vereinen mit einem Fokus auf (Amateur-)Sportvereine, reflektiert domänenspezifische Anwendungsszenarien und rekonstruiert dazu Geschäftsprozesse, Dokumente und Datenhaltung in Vereinen und entwickelt darauf aufbauend Anforderungen an die zu entwickelnde Webapplikation.

Besonderheit der Forschung durch Entwicklung an Sportyweb ist der akademische Software-Entwicklungsprozess, der öffentlich unter einer Open-Source-Software-Lizenz (**AGPL3**) erfolgt: https://github.com/Lehrstuhl-BWL-EvIS/sportyweb.

Mit der funktionalen Programmiersprache Elixir, dem Webapplikationsframework Phoenix und dem Datenbankmanagementsystem PostgreSQL kommen State-of-the-Art-Technologien moderner Webentwicklung zum Einsatz.

The SPORT research project develops and researches the web application Sportyweb – an integrated software application for sports clubs. The aim of the research project is the reflective development of a multitenancy web application for the efficient and collaborative management of sports clubs. Sportyweb targets a broad range of sports clubs of sizes from large-sized, multi-sports clubs with several

thousand members to small, single-sports clubs with a few dozen members. The research project examines the domain of clubs, in particular sports clubs, which has so far only been given secondary consideration, reflects domain-specific use cases, business processes and corresponding data management and develops requirements for the web application.

The research on Sportyweb follows the tenets of research through development (or construction-oriented research) based on an academic software development process carried out publicly under an open source software licence (**AGPL3**): **https://github.com/Lehrstuhl-BWL-EvIS/sportyweb**.

With the functional programming language **Elixir**, the web application framework **Phoenix** and the database management system **PostgreSQL**, state-of-the-art technologies of modern web development are used.

Motivation

Sports clubs are of great importance to modern society. They promote the physical and mental health and fitness of people into old age, help to build friendships, create cohesion and a sense of community, support the development of discipline and self-confidence in individuals and contribute to socialization and integration.

They cover a wide range of different sports. These range from popular sports (soccer, gymnastics, tennis, shooting, athletics, swimming, cycling, etc.) to niche sports (curling, disc golf, ice climbing, paragliding, etc.), with clubs either specializing in individual sports or combining several under one roof as multi-sport clubs.

The average German sports club has a few hundred members, but several thousand members are not uncommon. This makes the management of such organizations complex and requires an administrative effort that should not be underestimated. Especially for amateur sports clubs, this is becoming – or already is – an ever-increasing problem due to a diverse and growing range of tasks, high and ever-increasing demands and various social changes.

The “SPORT” research project was created to find a solution to at least some of these challenges. The aim is to develop the web application “Sportyweb” – an integrated software that enables the efficient and collaborative organization and administration of multi-sports clubs, some of which have many thousands of members.

The software is initially aimed at club officials (management and administration, department and group leaders, trainers) and is intended to support them in their respective tasks. Various facets of membership management (master data management, management of membership contracts, assignment to departments and groups), the planning, organization, implementation and administration of events, the management and rental of locations and equipment as well as the area of finance and accounting play central roles.

If you want to read more about the research project, have a look at this article (in German): **https://www.fernuni-hagen.de/universitaet/aktuelles/2024/07/olympia-special-software-fuer-vereine.shtml**

Research objectives

The research objectives of the project are diverse and cover a wide range of questions relating to the domain of sports clubs and the design and development of software.

- **Domain "sports clubs"**
  - Status of digitalization in (sports) clubs, readiness for change?
  - Technical requirements for administrative software for sports clubs?
  - Existing business processes and their differences, possible standardization, digitalization, automation?
- **Modeling**
  - How can a modular software architecture be achieved despite the close integration of the application's components?
  - Requirements for the data storage structures?
  - Derivation of generic reference models (reference data models and reference process models) of integrated software systems?
- **Software design & implementation**
  - Functional and non-functional requirements for administrative software for sports clubs?
  - Challenges in the practical implementation of recommendations and best practices from the field of IT security?
  - Challenges in the practical implementation of legal data protection requirements?
- **User experience & user interface design**
  - How to design an intuitive user experience despite the high complexity of the domain?
  - How to design user-friendly interfaces that can be understood even without extensive IT knowledge?
- **Approaches to modern web application development**
  - Scalable, performant and fault-tolerant web applications with the functional programming language **Elixir**, the web application framework **Phoenix**, and the database management system **PostgreSQL**?
  - Successful software development process despite a high fluctuation in an academic environment?

Project repository

https://github.com/Lehrstuhl-BWL-EvIS/sportyweb

Join us & contribute!

There are various ways to participate in the SPORT research project:

- For students of the FernUniversität in Hagen, we offer topic suggestions for Bachelor's and Master's theses: **https://github.com/Lehrstuhl-BWL-EvIS/abschlussarbeiten** (You are also welcome to contact us with your own topic suggestions!)
- We look forward to ideas, feature requests, suggestions and feedback from club officials.
- Anyone who finds bugs when using Sportyweb can create issues in the project repository on GitHub or submit fixes via pull requests.

Your advantages of joining us

Working on the SPORT research project in the form of a thesis is intensive and demanding! And yet it is worth it, as the results of your work will not gather dust in a drawer after submission and evaluation, but get integrated into an open source project that creates value for society in the long term.

Other advantages at a glance:

- You will learn the advanced conception and design of a comprehensive software development project and the associated processes in a practical way.
- You will acquire in-depth, practical skills in the development of information systems, especially web applications, which you can easily transfer to other technology stacks over the course of your future career.
- You will familiarise yourself with the functional programming paradigm to discover new ways of thinking and different approaches, which are increasingly finding their way into object-oriented programming languages such as Java, C# or Python.
- You will be involved in an open source project that you can refer to on your CV, giving you a strategic advantage over your competitors.
- You will lay the foundations for a career as a software developer, software architect, **product owner**, engineering manager or IT project manager.
- If you are planning to start a company in the IT sector, you will acquire the technical knowledge you need to turn your ideas into reality.

You will be given enough time, literature and further information to familiarise yourself efficiently with the technologies used before the official start of your thesis!

Requirements for participation

The most important prerequisite for joining the research project is the willingness to learn about new technologies and to close possible gaps in your knowledge!

For Bachelor's theses:

- Basic knowledge of software engineering & software development (regardless of any specific programming language)

For Master's theses:

- Advanced knowledge of software engineering & software development (regardless of any specific programming language)
- Basic knowledge of web development (HTML, CSS, ...)
- Basic knowledge of data modelling and database development (SQL)
- Basic knowledge of UX- & UI-Design
- Interest in the development of web applications

Ideally, you were or are active in a club – it does not have to be a sports club – or are willing to familiarise yourself with the domain of clubs.

Contact

If you are interested in working with us, have questions about the research project, want to know more about open thesis topics or have feedback, please contact **Prof. Dr. Stefan Strecker** and **Bastian Kres** by e-mail.

Publications, Theses, Working Papers

- Strecker S (2021–2024) ***Überlegungen zu Anforderungen an eine integrierte Anwendungssoftware für Sportvereine und Entwurf eines initialen Datenmodells.***
- Rühe S (2024) *Entwurf, Entwicklung und Evaluation der Softwarebibliothek „ex_invoice“ zur Erzeugung rechtskonformer Rechnungen im PDF-Format mit der Programmiersprache Elixir.* Bachelorarbeit im Studiengang B.Sc. Wirtschaftsinformatik.
- Göttsching S (2024) *Entwurf, Entwicklung und Evaluation des Softwarepaketes „ex_sepa“ zur Erzeugung von SEPA-XML-Daten mit der Programmiersprache Elixir.* Bachelorarbeit im Studiengang B.Sc. Wirtschaftsinformatik.
- Kres B (2024) *Entwurf, Implementierung und Evaluation einer integrierten Anwendungssoftware für Sportvereine als Webapplikation auf Basis von Elixir & Phoenix.* Masterarbeit im Studiengang M.Sc. Wirtschaftsinformatik.
- Utley A (2023) *Entwurf, Implementierung und Evaluation eines Subsystems zur*

Authentifizierung, Autorisierung und Zugriffskontrolle als Teil einer auf Elixir & Phoenix basierenden Webapplikation für Sportvereine. Masterarbeit im Studiengang M.Sc. Wirtschaftswissenschaft für Ingenieur/-innen und Naturwissenschaftler/-innen.

- Fava L (2022) *Datenbankentwicklung für eine WebApp für Amateursportvereine mit PostgreSQL.* Masterarbeit im Studiengang M.Sc. Wirtschaftsinformatik.
- Farzankia S (2022) *SPORT - Forschung und Entwicklung von Sportyweb. Entwicklung eines Demonstrationsprototyps der Benutzungsschnittstelle.* Masterarbeit im Studiengang M.Sc. Wirtschaftsinformatik.
- Geiling D (2022) *Prototypische Implementierung einer Web-Applikation für Amateursportvereine mit Elixir/Phoenix.* Bachelorarbeit im Studiengang B.Sc. Wirtschaftsinformatik.
- Espey K (2022) *Entwurf eines Mock-Ups für die Benutzungsschnittstelle einer integrierten Web-Applikation für Amateursportvereine.* Bachelorarbeit im Studiengang B.Sc. Wirtschaftsinformatik.
- Halscheidt S (2021) *Entwurf eines umfassenden Datenmodells für eine integrierte Anwendungssoftware für Sportvereine.* Masterarbeit im Studiengang M.Sc. Wirtschaftsinformatik.
- Baierl J (2021) *Anforderungsanalyse für eine Verwaltungssoftware von Sportvereinen unter Berücksichtigung der Methoden der agilen Softwareentwicklung.* Masterarbeit im Studiengang M.Sc. Wirtschaftswissenschaft.